BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SAHIWAL

INTERMEDIATE (12TH CLASS) EXAMINATION, 2021

CHEMISTRY

Past Paper 2021 • Paper Code: 8483

Retyped faithfully from the original board paper — same questions, no changes.

Chemistry - Paper: II - Objective (II)

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 8483

Note: Note: - You have four choices for each objective type question as A, B, C and D. The choice which you think is correct; fill that circle in front of that question number in your answer book. Use marker or pen to fill the circles. Cutting or filling up two or more circles will result no mark.

Note: Objective - (II)

Q1. Which reagent is used to reduce a carboxylic group to an alcohol?

(A) H_2/Ni
(B) H_2/Pt
(C) $NaBH_4$
(D) $LiAlH_4$

Q2. Which of the given will have the highest boiling point?

(A) Methanal
(B) Ethanal
(C) Propanal
(D) 2-Hexanone

Q3. Elimination bimolecular reactions involve:

(A) first order kinetics
(B) second order kinetics
(C) third order kinetics
(D) zero order kinetics

Q4. Which compound will have maximum repulsion with H_2O?

(A) C_6H_6
(B) C_2H_5OH
(C) $CH_3CH_2CH_2OH$
(D) CH_3-O-CH_3

Q5. Which of the given compound will react with Tollen's reagent?

(A) CH_3-C(=O)-H
(B) CH_3-C(=O)-CH_3
(C) CH_3-C(=O)-OH
(D) CH_3-C(=O)-CH_2-CH_3

Q6. Which compound is the most reactive one?

(A) Benzene
(B) Ethene
(C) Ethane
(D) Ethyne

Q7. The presence of a double bond in a compound is the sign of:

(A) saturation
(B) unsaturation
(C) substitution
(D) none

Q8. A double bond consists of:

(A) two sigma bonds
(B) one sigma and one pi-bond
(C) one sigma two pi-bonds
(D) two pi-bonds

Q9. The strength of binding energy of transition elements depends upon number of:

(A) electron pairs
(B) unpaired electrons
(C) neutrons
(D) protons

Q10. Laughing gas is chemically:

(A) NO
(B) N_2O
(C) NO_2
(D) N_2O_4

Q11. Which is the strongest acid?

(A) HClO
(B) $HClO_2$
(C) $HClO_3$
(D) $HClO_4$

Q12. Which of the given element is not present abundantly in earth's crust?

(A) Silicon
(B) Aluminium
(C) Sodium
(D) Oxygen

Q13. Chile saltpetre has the chemical formula:

(A) $NaNO_3$
(B) KNO_3
(C) $Na_2B_4O_7$
(D) $Na_2CO_3.H_2O$

Q14. The correct statement is:

(A) Na^+ is smaller than Na atom
(B) Na^+ is larger than Na atom
(C) Cl^- is smaller than Cl atom
(D) Cl^- (ion) and Cl (atom) are equal in size

Q15. The chemical formula of Fluorospar is:

(A) $Ca_3(PO_4)_2$
(B) CaF_2
(C) Na_3AlF_6
(D) $KCl.MgCl_2.6H_2O$

Q16. Phosphorus helps the growth of:

(A) roots
(B) leaves
(C) stem
(D) seed

Q17. Acetic acid is manufactured by:

(A) Distillation
(B) Fermentation
(C) Ozonolysis
(D) Esterification

Chemistry - Paper: II - Subjective

Time Allowed: 2:40 Hours | **Maximum Marks:** 68

Note: Note: Section I is compulsory. Attempt any 3 questions from Section II.

Note: (SECTION-I)

Q2.. (Marks 16) Write short answers to any Eight parts. (8 x 2 = 16)

i. What is trend of electron affinity from top to bottom in a group?

ii. Why the ionic radii of negative ions are larger than the size of their parent atom?

iii. Give the advantages of Down's cell for the preparation of sodium metal on commercial scale.

iv. BeO is amphoteric in nature. Justify.

v. What is borax bead test?

vi. Write down any two uses of boric acid.

vii. Give any four uses of Aluminium.

viii. Write down the dissimilarities of oxygen with sulphur (any four).

ix. Why SO_3 dissolves in H_2SO_4, not in water?

x. Why nitrogen is important for plants? Give two name of nitrogeneous fertilizers.

xi. Write the reactions involved in preparation of urea fertilizer.

xii. Write down the reaction which takes place in 24 hours during setting of cement.

Q3.. (Marks 16) Write short answers to any Eight parts. (8 x 2 = 16)

i. Why HF is weak acid?

ii. Write reactions of chlorine with cold and hot NaOH.

iii. What is meant by available chlorine in bleaching powder? Give reaction.

iv. Define Corrosion.

v. Why compounds of transition elements show colour?

vi. Define Monocyclic and Polycyclic aromatic hydrocarbons.

vii. Define Resonance Energy. Give one example.

viii. Write the reactions of formaldehyde and acetaldehyde with HCN

ix. Write applications of Iodoform test.

x. How acetic acid is prepared from Grignard's reagent?

xi. Write the reactions of Acetic Acid with NaOH and Na_2CO_3.

xii. Write four uses of Acetic Acid.

Q4.. (Marks 12) Write short answers to any Six parts. (6 x 2 = 12)

i. What is catalytic cracking?

ii. What is meant by functional group? Write the name of two oxygen containing functional groups.

iii. Alkanes are less reactive than Alkenes. Justify.

iv. Define Hydroxylation. Give an example.

v. How may Ethene be converted into ethyl alcohol?

vi. Define Alkyl Halide. Which is the best method of preparing alkyl halides?

vii. Write IUPAC names of the following compounds: (a) $(CH_3CH_2)_3CBr$ (b) C_2H_5-$CH(CH_3)$-$CH(C_2H_5)$-Br

viii. How primary, secondary and tertiary alcohols are different from each other in structure?

ix. Why ethanol has higher boiling point than diethyl ether but less than water?

> Note: (SECTION-II) (Each question carries Eight (8) Marks)

Q5.. (a) Justify the position of hydrogen at the top of group I-A by giving any four points of similarities and dissimilarities. (b) How Lithium shows peculiar behaviour among alkali metals? Mention any eight properties.

Q6.. (a) How does Conc. HNO_3 react with the following metals? (i) Cu (ii) Hg (iii) Sn (iv) Zn (b) Explain Electrochemical theory of corrosion.

Q7.. (a) Write four various forms of structural isomerism with examples. (b) Explain Aldol Condensation with mechanism using a suitable example.

Q8.. (a) Name the following compounds according to IUPAC system: (i) $(CH_3)_3C{\equiv}CH$ (ii) $(CH_3CH_2)_2CH$ (iii) $HC{\equiv}C\text{-}CH{=}CH\text{-}CH_3$ (iv) $HC{\equiv}C\text{-}CH(CH_3)\text{-}CH{=}CH_2$ [best-effort] (b) Give four equations with conditions for the preparation of Alkyl halides from Alcohols.

Q9.. (a) Write Friedel-Craft Alkylation reaction with mechanism. (b) Starting from phenol prepare the following compounds: (i) Ortho Hydroxybenzyl Alcohol (ii) Para Hydroxybenzyl Alcohol (iii) Bakelite